Text as Data

Class 4: Dictionaries, annotation, and validation

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Where we are in the course

- Class 1: text becomes documents, tokens, and vectors
- Class 2: counts, rates, log odds, and preprocessing choices
- Class 3: a small authorship case study with the Federalist Papers
- Today: turn a concept into a transparent, checkable text measure

The problem is measurement

We often want to measure an idea that is not itself a word: attention to immigration, policy frames, incivility, or employment.

Text gives us traces of those ideas. A dictionary is a rule for turning traces into an observed label or score.

The question is not simply "which words occur?" It is "does this rule measure the concept we claim it measures?"

A word is not a concept

Sentence	Does it discuss employment?
"The factory closure will eliminate 600 local jobs."	Yes
"This small engine does the job in under a second."	No
"The recession left many people without work for months."	Yes, even though it contains neither `job` nor `employment`

Keyword matching is an appealing starting point. It is rarely a complete measurement rule.

From concept to observed data



Every arrow is a modelling choice. We can inspect and revise those choices.

Start with a codebook

A codebook tells a reader how the concept will be identified before looking at results.

For each label, specify:

- the unit of analysis: document, paragraph, sentence, or post
- what counts as an instance of the concept
- what does not count, including likely ambiguous cases
- how context, negation, uncertainty, and mixed cases are handled

A small codebook: employment attention

Unit	Sentence
Include	Jobs, employment, unemployment, hiring, layoffs, vacancies, wages, hours, employers, and employment-related training
Exclude	Technical jobs, a job as a task, generic effort, and *employ* meaning use
Context	Negation changes the claim but not the topic. Use the full sentence to disambiguate terms.

The full teaching codebook is in `CODEBOOK.md`.

Define the unit before you count

Document level

"Does this State of the Union address discuss employment?"

Useful for broad agendas, but one mention can label a long document.

Sentence level

"Does this sentence discuss employment?"

Useful for validation and close error inspection, but requires more labels.

The unit determines what a false positive or false negative means.

A dictionary is an operational rule

The simplest dictionary marks a unit if it contains at least one term from a list.

$$\text{label}(d) = \mathbb{1}\{\texttt{job}, \texttt{jobs}, \texttt{employment}, \texttt{unemployment} \in d\}.$$

This is easy to reproduce and explain. It also misses paraphrases and can count irrelevant uses of the same word.

Matching details change the measure

Rule	Matches	Does not match
`job`	job, jobs, jobless (substring)	jobless with word boundaries
`\bjobs?\b`	job, jobs	jobless, jobsworth
`\bunemploy\w*`	unemployment, unemployed	employment
`\bhir(e|ing|ed)\b`	hire, hiring, hired	rehire unless explicitly added

Regular expressions are useful because they make the matching rule explicit. They do not solve ambiguity on their own.

A formal dictionary count

Let $d = (w_1, \ldots, w_n)$ be a text of n tokens and D a dictionary of terms.

$$c_D(d) = \sum_{i=1}^{n} \mathbf{1}\{w_i \in D\} = \sum_{w \in D} \text{count}(w, d).$$

The indicator $\mathbf{1}\{\cdot\}$ equals 1 when the condition is true and 0 otherwise. Each matching token contributes one to the count.

For phrase or regex dictionaries, count qualifying pattern matches instead of exact token matches. Specify how variants and overlapping matches are handled.

A dictionary classification rule

Apply a threshold τ to the dictionary count:

$$\hat{y}_D(d;\tau) = \mathbf{1}\{c_D(d) \geq \tau\} = \begin{cases} 1 & \text{if } c_D(d) \geq \tau, \\ 0 & \text{otherwise.} \end{cases}$$

With $\tau = 1$, a sentence is labelled positive when it contains at least one qualifying dictionary match.

The count, matching rule, and threshold together define the classifier. Raising the threshold changes which texts receive the positive label.

Continuous dictionary scores

Keep the raw count or divide it by the number of tokens:

$$r_D(d) = \frac{c_D(d)}{n}.$$

For a weighted dictionary, give each term a weight a_w, with $a_w = 0$ for words outside the dictionary:

$$s_D(d) = \sum_{i=1}^{n} a_{w_i} = \sum_{w \in D} a_w \, \text{count}(w, d).$$

Weights can express direction or intensity, as in sentiment measurement.

A continuous score can be used as a variable in a regression. Computing the dictionary score itself does not fit a regression model.

Positive-negative sentiment classification

Let D_+ and D_- be positive and negative term lists. Assign weights $+1$ and -1:

$$s_{\text{sent}}(d) = c_{D_+}(d) - c_{D_-}(d), \qquad \hat{y}_{\text{sent}}(d) = \mathbf{1}\{s_{\text{sent}}(d) > 0\}.$$

Example: "a **wonderful** but **boring** film"

$$c_{D_+}(d) = 1, \quad c_{D_-}(d) = 1, \quad s_{\text{sent}}(d) = 0.$$

The notebook assigns a zero score to the negative class, so this example receives label 0.

Word order and contrast are absent from this rule. Error inspection shows what that simplification costs.

Rates answer a different question than counts

Speech	Employment terms	Tokens	Rate per 1,000 tokens
A	8	1,000	8.0
B	10	2,500	4.0

Speech B contains more matched terms, but employment language is less concentrated. Which measure is appropriate depends on whether we care about total attention or rhetorical emphasis.

Existing dictionaries can be useful starting points

- They make prior measurement decisions visible.
- They can save time and support comparison across studies.
- Their vocabulary may reflect a different period, genre, country, or language.
- They still need to be checked in the text and corpus at hand.

Reusing a dictionary transfers a measurement claim. It does not validate it automatically.

For an overview of dictionary methods, see Grimmer, Roberts, and Stewart (2022), *Text as Data*.

A search string defines the corpus

Before we classify or score texts, a search query may decide which texts enter the analysis:

$$\mathcal{C}_q = \{d : d \text{ matches query } q\}.$$

- A narrow query can omit relevant texts, reducing coverage.
- A broad query can retrieve irrelevant texts, adding noise.
- The retrieved set may not represent the discourse or population we want to describe.

Treat the search string as a measurement and sampling decision. Check retrieved and missed examples before making claims about the corpus.

Why validation is necessary

Our dictionary will make two kinds of mistakes:

- **False positive:** it labels a unit as employment even though the human codebook says it is not.
- **False negative:** it misses a unit that a human coder judges to be about employment.

The mistake patterns tell us what the rule is actually measuring.

Human annotation creates a benchmark

1. Write the codebook.
2. Label a sample of texts while following it.
3. Resolve unclear cases and revise the codebook if needed.
4. Keep a held-out set for evaluating the final rule.

Human labels are not magic truth. They are a transparent reference based on the same substantive definition.

Building a credible manual baseline

1. Draft the codebook from the research question and pilot examples.
2. Train at least two coders on a shared subset, then inspect disagreement and revise unclear rules.
3. Double-code a subset to assess agreement. Agreement is a diagnostic, not proof that the labels are valid.
4. Freeze the codebook, label a development set, and reserve a representative held-out set for final evaluation.

When possible, keep dictionary development and final validation separate, either across people or across samples. This reduces the temptation to tune rules to the answer already seen.

Annotation is also measurement

Two careful people can disagree because the concept, unit, or decision rule is underspecified.

Disagreement is evidence. Read the disputed passages, decide what the codebook should say, and apply the rule consistently.

The goal is not to eliminate judgement. It is to document and discipline it.

See Denny and Spirling (2018) on measurement choices in text analysis.

Compare the rule with the reference labels

		Human reference	
		Employment	Not employment
Dictionary	Match	true positive	false positive
	No match	false negative	true negative

Read examples in every cell. A single summary statistic cannot show where the rule fails.

A useful hypothesis-testing analogy

Confusion-matrix quantity	What it asks	Hypothesis-testing analogue
False-positive rate $\frac{FP}{FP+TN}$	Among passages that are not about employment, how often does the rule label them positive?	Type I error rate
False-negative rate $\frac{FN}{FN+TP}$	Among employment passages, how often does the rule miss them?	Type II error rate

The analogy has limits. An α level is chosen before a statistical test to control Type I errors under a null model. Here, we estimate error rates by comparing a text rule with human reference labels. There is no null distribution or p-value.

Precision, recall, and F1

$$\text{precision} = \frac{TP}{TP + FP} \qquad \text{recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Precision asks: among labelled passages, how many are actually about employment? Recall asks: among employment passages, how many did we find? F1 combines the two but does not replace looking at errors.

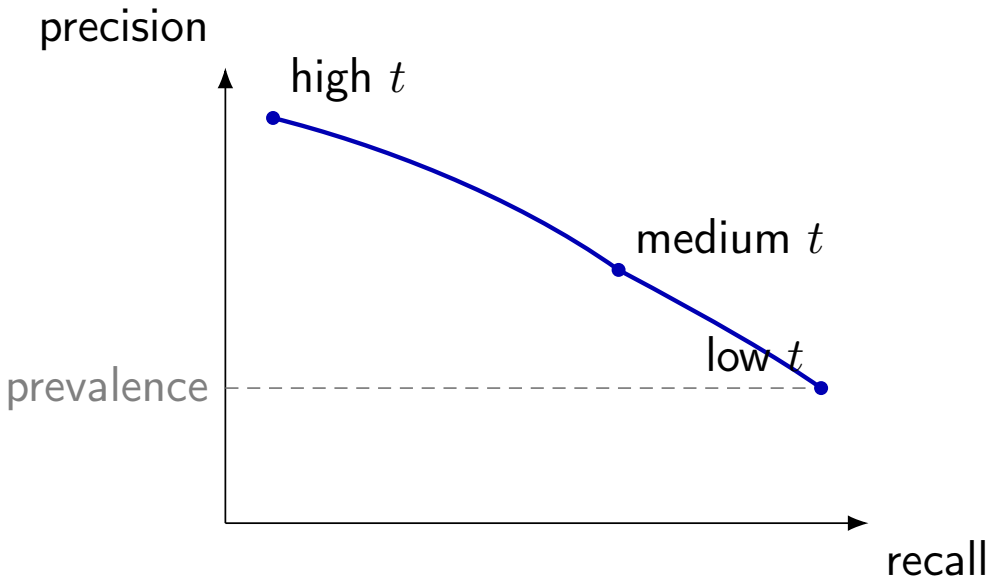
Precision and recall depend on a threshold

Some text methods produce a score or ranking rather than a fixed yes/no label. Choose a threshold t and label a passage positive when its score is at least t.

Threshold	Precision	Recall	Interpretation
High	high	low	label only the clearest passages
Medium	moderate	moderate	include plausible passages
Low	low	high	cast a wider net

For a dictionary, the score might be the number of matches. For a classifier, it is often a predicted probability or another model score.

A precision-recall curve



Each point is a different threshold on the same scored texts.

Moving toward higher recall usually includes more false positives, so precision often falls.

The curve shows the available trade-offs before choosing an operating threshold.

Why use a PR curve?

Precision-recall curves are especially useful when the positive class is rare, such as a narrow policy frame or a particular form of incivility.

- The horizontal reference is the prevalence of the positive label in the validation set.
- A useful method stays above that baseline and provides a good precision–recall trade-off where the research design needs it.
- Average precision summarises a ranked curve, but it does not select the threshold or reveal the error cases.

Report the validation set, the curve or a table of thresholds, and the threshold used in the analysis.

Accuracy can mislead with a rare topic

Suppose only 5 of 100 sentences discuss employment. A rule that always says "not employment" is 95% accurate.

It also has zero recall and does not measure employment attention at all.

Report the class balance, precision, recall, and examples of errors.

Error inspection drives revision

Observed error	Possible response
Technical "job" is matched	Require a work or labour-market context, or exclude clear technical phrases
"Without work" is missed	Add a phrase rule based on annotated examples
One named appointment is matched	Distinguish generic labour-market hiring from a single officeholder

Revise the dictionary because the codebook supports a change, not because a single result is inconvenient.

Do not test on the same examples forever

If we repeatedly tune a dictionary to the same labelled passages, the apparent score will become too optimistic.

Use one set of annotated texts to develop the rule and another set to evaluate it.

For a small classroom example, this is a habit to practice. For a study, it is part of credible validation.

What to report in a paper

- The concept, unit, and codebook
- The dictionary or matching rules, ideally in an appendix or repository
- How preprocessing was done
- The score used: binary label, count, rate, or another transformation
- Validation design, error patterns, and limits of the measure

Transparency lets readers disagree with a choice without guessing what the choice was.

Practical: build and evaluate a small dictionary

The notebook includes twenty invented sentences and the codebook used to label them.

You will:

1. run a basic keyword dictionary
2. inspect its false positives and false negatives
3. add explicit regular-expression rules
4. calculate precision, recall, and F1 against the reference labels

Second practical: sentiment dictionary classification

The notebook also includes 10,662 Rotten Tomatoes review snippets from the Pang and Lee sentence-polarity dataset: 5,331 from positive reviews and 5,331 from negative reviews.

Build a simple positive–negative dictionary score, then classify a snippet as positive when its positive terms outweigh its negative terms.

The labels are inherited from each source review's *fresh* or *rotten* status. They are a benchmark for sentiment classification, not direct human labels for every snippet.

Pang and Lee (2005), "Seeing Stars: Exploiting Class Relationships for Sentiment Categorization with Respect to Rating Scales."

A substantive question for the practical

The first dictionary counts only a short list of terms. After inspecting its errors, decide whether you want to prioritize:

- **precision**, by marking only unambiguous employment language, or
- **recall**, by adding broader vocabulary and then checking more false positives.

There is no universal best choice. The appropriate trade-off follows from the research question and the cost of each error.

Key takeaways

- Dictionaries are measurement models, not just word lists.
- Codebooks make the substantive rule explicit.
- Matching details, units, and normalisation change the data we create.
- Validation requires human reference labels and direct inspection of errors.
- Next: supervised classification learns a rule from annotated examples.

Sources

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